

## DIGITAL IMAGE PROCESSING LABORATORY

|                                    |                          |
|------------------------------------|--------------------------|
| <b>Semester</b>                    | B.E. Semester VII        |
| <b>Subject</b>                     | Digital Image Processing |
| <b>Subject Professor In-charge</b> | Prof. Rajashree Soman    |
| <b>Laboratory</b>                  | LO5                      |

|                                              |            |                 |   |
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| <b>Experiment Number</b>              | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Experiment Title</b>               | Region Growing, Connectivity Comparison, and Region Splitting and Merging                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Aim</b>                            | <p>To implement seeded region growing (with the region mean updated as pixels are absorbed) for segmenting the iris/pupil in grayscale near-infrared eye-crop images; to compare 4-neighbourhood and 8-neighbourhood connectivity quantitatively (region size, shape, boundary behaviour and processing time); and to implement quadtree-based region splitting and merging as a second region-based segmentation technique – continuing the enhancement (Task 3, histogram equalization) → edge/threshold segmentation (Task 6) pipeline on the same 3 dataset images, with quantitative rather than visual-only evaluation throughout.</p> <p><b>Why the pupil is the region of interest.</b> In every one of these near-IR eye crops the pupil/iris is the darkest, most compact, roughly circular structure against a brighter sclera and skin – exactly the seed-and-grow shape assumption region growing depends on, and its area over time is the geometric signal PERCLOS-style drowsiness detection (<code>src/perclos.py</code>) ultimately consumes: a shrinking or vanishing dark region is the visible eye-closure signal this whole project is built to detect.</p>                                                                                    |
| <b>Resources / Apparatus Required</b> | Software: Python 3.11 • OpenCV • NumPy • Matplotlib • scikit-image • Pillow • Jupyter Notebook / VS Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Algorithm</b>                      | <p><b>A. Region Growing</b></p> <ol style="list-style-type: none"> <li>1. Start the program.</li> <li>2. Read the enhanced (histogram-equalized) grayscale eye-crop image as a NumPy <code>uint8</code> array (same enhancement as Task 6's input, for pipeline continuity).</li> <li>3. Compute an <b>automatic seed</b>: Gaussian-blur the image (5x5) to suppress single-pixel noise, then take the coordinate of the blurred image's global minimum (darkest point) – this reliably lands inside the pupil in all 3 test images.</li> <li>4. Also record a <b>manual seed</b>: the geometric centre of the crop, for comparison.</li> <li>5. From the seed, run BFS: maintain a running region mean (initially the seed's own intensity) and a visited set; for every unvisited 4- or 8-neighbour of a pixel just absorbed, admit it into the region if <code>abs(neighbour - running_mean) &lt;= T</code>.</li> <li>6. <b>Update the region mean</b> after every pixel is absorbed (running mean over all pixels absorbed so far), so later admission decisions track the region's own evolving statistics rather than only the original seed value.</li> <li>7. Repeat until the queue is empty; output the binary region mask and its pixel count.</li> </ol> |

8. Repeat for 3 thresholds (Low, Medium, High) and, for the metrics table, a 4th (Very High) to locate where the region starts leaking past the expected pupil area.

9. Stop the program.

#### B. 4- vs 8-Connectivity

1. Start the program.
2. Using the same auto seed and a fixed threshold (Medium,  $T=25$ ), run region growing twice on the same image: once considering only the 4 orthogonal neighbours (up/down/left/right), once considering all 8 neighbours (orthogonal + 4 diagonals).
3. Compute the pixel-count difference, the symmetric difference (XOR) mask between the two results, and each mask's shape compactness ( $4\pi \cdot \text{Area} / \text{Perimeter}^2$  on the largest external contour, via `cv2.findContours / cv2.contourArea / cv2.arcLength`).
4. Time both variants over 30 repeated calls on the same image/seed/threshold and report the mean per-call time in milliseconds.
5. Stop the program.

#### C. Region Splitting and Merging

1. Start the program.
2. Recursively **split** the image as a quadtree: starting from the whole frame, test a block's homogeneity as `max(block) - min(block) <= threshold`; if homogeneous, or the block has hit the minimum block size, keep it as a leaf; otherwise split it into 4 quadrants (top-left/top-right/bottom-left/bottom-right) and recurse on each.
3. Compute each leaf's mean intensity.
4. **Merge**: build an adjacency graph over leaves that share a boundary edge; union two adjacent leaves (via union-find) whenever their mean intensities differ by no more than the same tolerance used for splitting.
5. Relabel every leaf by its union-find root to produce the final merged-region label map.
6. Report the number of leaves after splitting and the number of regions remaining after merging.
7. Stop the program.

#### Program code

Trimmed to the essential logic. Full module: `src/region_growing.py`. Every function takes/returns a single-channel `uint8 NumPy` array; nothing in `src/enhancement.py` or `src/segment.py` (both reused as-is) is modified.

#### A. Region Growing

```
def auto_seed(img, blur_ksize=5):
    blurred = cv2.GaussianBlur(img, (blur_ksize, blur_ksize), 0)
    idx = int(np.argmax(blurred))
    return np.unravel_index(idx, blurred.shape) # (row, col)

def region_grow(img, seed, T, connectivity=8, update_mean=True):
    neighbours = CONN8 if connectivity == 8 else CONN4
    visited = np.zeros(img.shape, dtype=bool); mask =
    np.zeros(img.shape, dtype=np.uint8)
    sr, sc = seed
    visited[sr, sc] = True; mask[sr, sc] = 255
    region_sum, region_count = float(img[sr, sc]), 1
    mean = region_sum / region_count
    q = deque([(sr, sc)])
    while q:
        r, c = q.popleft()
        for dr, dc in neighbours:
            nr, nc = r + dr, c + dc
            if 0 <= nr < h and 0 <= nc < w and not visited[nr, nc]:
                visited[nr, nc] = True
```

```

        if abs(int(img[nr, nc]) - mean) <= T:
            mask[nr, nc] = 255
            region_count += 1; region_sum += img[nr, nc]
            if update_mean:
                # "update the region
                mean if required"
                mean = region_sum / region_count
            q.append((nr, nc))
        return mask, region_count

```

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### C. Region Splitting and Merging

```

def _homogeneous(block, threshold):
    return int(block.max()) - int(block.min()) <= threshold

def _quadtree_split(img, r0, r1, c0, c1, min_size, threshold, leaves):
    h, w = r1 - r0, c1 - c0
    if h <= min_size or w <= min_size or _homogeneous(img[r0:r1, c0:c1],
    threshold):
        leaves.append((r0, r1, c0, c1)); return
    rm, cm = r0 + h // 2, c0 + w // 2
    _quadtree_split(img, r0, rm, c0, cm, min_size, threshold, leaves) #
    top-left
    _quadtree_split(img, r0, rm, cm, c1, min_size, threshold, leaves) #
    top-right
    _quadtree_split(img, rm, r1, c0, cm, min_size, threshold, leaves) #
    bottom-left
    _quadtree_split(img, rm, r1, cm, c1, min_size, threshold, leaves) #
    bottom-right

def split_merge(img, min_size=20, homogeneity_threshold=25,
merge_tolerance=None):
    leaves = []; _quadtree_split(img, 0, h, 0, w, min_size,
    homogeneity_threshold, leaves)
    means = [float(img[r0:r1, c0:c1].mean()) for (r0,r1,c0,c1) in
    leaves]
    parent = list(range(len(leaves))) # union-find
    for i, j in adjacent_leaf_pairs(leaves):
        if abs(means[i] - means[j]) <= (merge_tolerance or
    homogeneity_threshold):
            union(parent, i, j)
    return relabel_by_root(leaves, parent) # -> labels,
    n_leaves, n_regions

```

### Output

Same 3 test images as Task 3 / Task 6, each loaded, upscaled with `cv2.INTER_NEAREST` (minimum 300px side), and histogram-equalized before region growing / splitting / merging.

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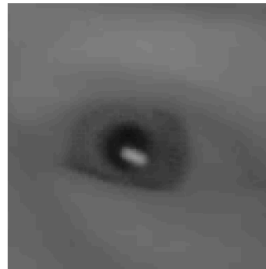
| Label         | File                               | Condition                                          | Native size | Native min / max intensit |
|---------------|------------------------------------|----------------------------------------------------|-------------|---------------------------|
| good_lighting | alert/s0031_00366_1_0_1_0_1_02.png | open eye, no glasses, good lighting, no reflection | 128x128     | 19 / 148                  |
| poor_lighting | alert/s0014_08297_0_0_1_1_0_02.png | open eye, no glasses, poor/uneven lighting         | 130x130     | 8 / 255                   |

|                    |                                    |                                                                            |         |          |
|--------------------|------------------------------------|----------------------------------------------------------------------------|---------|----------|
| glasses_reflection | alert/s0035_00133_0_1_1_2_1_02.png | open eye,<br>eyeglasses +<br>strong IR<br>reflection<br>(glare<br>hotspot) | 144x144 | 27 / 255 |
|--------------------|------------------------------------|----------------------------------------------------------------------------|---------|----------|

### Step 1 – original / grayscale, with dimensions and intensity range

Figure 1: Step 1 – good\_lighting: 128x128 native, min=19, max=148 (source file is already single-channel IR, so panels are numerically identical)

Original (as loaded, IMREAD\_COLOR)



Grayscale (cvtColor\_BGR2GRAY)

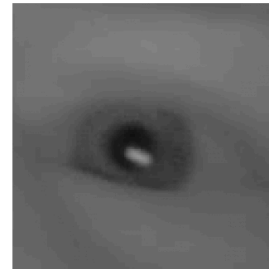


Figure 1: Step 1 – good\_lighting: 128x128, min=19, max=148.

Figure 2: Step 1 – poor\_lighting: 130x130 native, min=8, max=255 (source file is already single-channel IR, so panels are numerically identical)

Original (as loaded, IMREAD\_COLOR)



Grayscale (cvtColor\_BGR2GRAY)

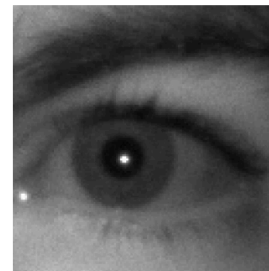
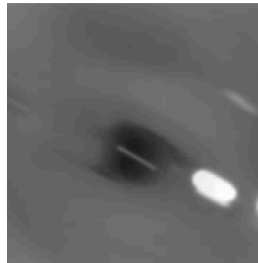


Figure 2: Step 1 – poor\_lighting: 130x130, min=8, max=255.

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Figure 3: Step 1 – glasses\_reflection: 144x144 native, min=27, max=255 (source file is already single-channel IR, so panels are numerically identical)

Original (as loaded, IMREAD\_COLOR)



Grayscale (cvtColor\_BGR2GRAY)

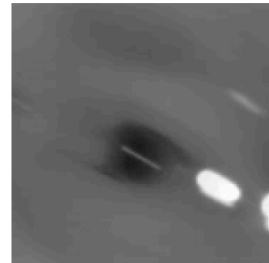


Figure 3: Step 1 – glasses\_reflection: 144x144, min=27, max=255.

The dataset PNGs are already single-channel infrared crops, so the "Original" (IMREAD\_COLOR, channel-replicated) and "Grayscale" (cvtColor\_BGR2GRAY) panels are numerically identical here – both panels are still produced to follow the standard Step-1 read/convert/report procedure.

### Seeds used

| Image | Auto seed<br>(row, col) | Auto seed<br>intensity | Manual seed<br>(row, col) | Manual seed<br>intensity |
|-------|-------------------------|------------------------|---------------------------|--------------------------|
|-------|-------------------------|------------------------|---------------------------|--------------------------|

|                    |            |   |                        |    |
|--------------------|------------|---|------------------------|----|
| good_lighting      | (123, 114) | 0 | (128, 128) –<br>centre | 1  |
| poor_lighting      | (133, 106) | 0 | (130, 130) –<br>centre | 54 |
| glasses_reflection | (152, 143) | 0 | (144, 144) –<br>centre | 3  |

The automatic seed (darkest point after a 5x5 Gaussian blur) lands exactly on a 0-intensity pixel inside the pupil on all 3 images. The manual seed (geometric centre of the crop) is close by construction but noticeably brighter on *poor\_lighting* (intensity 54 vs. 0) – the pupil there is off-centre in the crop, so a naive "assume the pupil is centred" manual placement is measurably worse than the automatic darkest-point seed.

### Region growing at 3 thresholds, per image

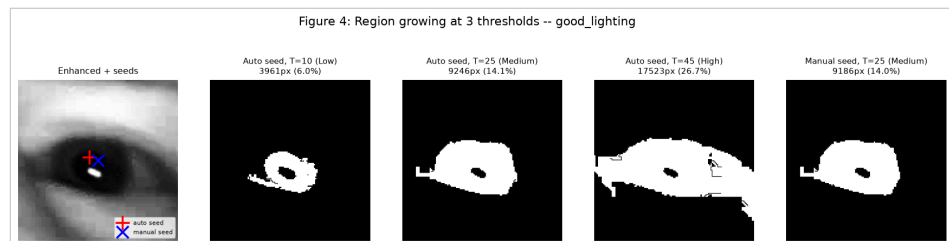


Figure 4: Region growing, auto seed at Low/Medium/High T, and manual seed at Medium T – good\_lighting.

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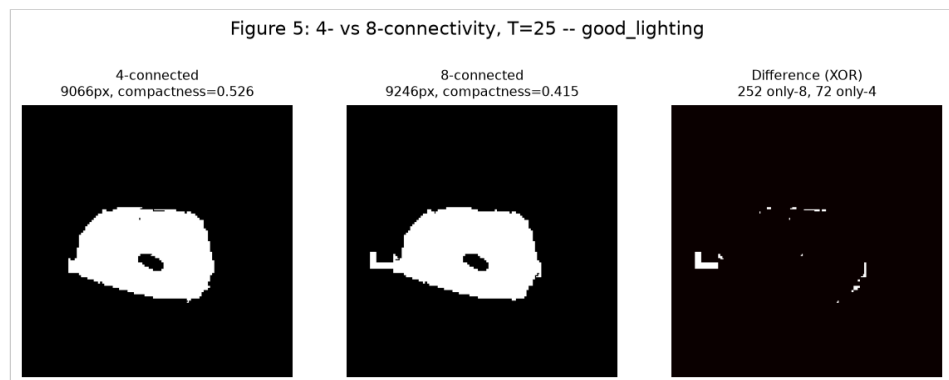


Figure 5: 4- vs 8-connectivity at T=25, with XOR difference map – good\_lighting.

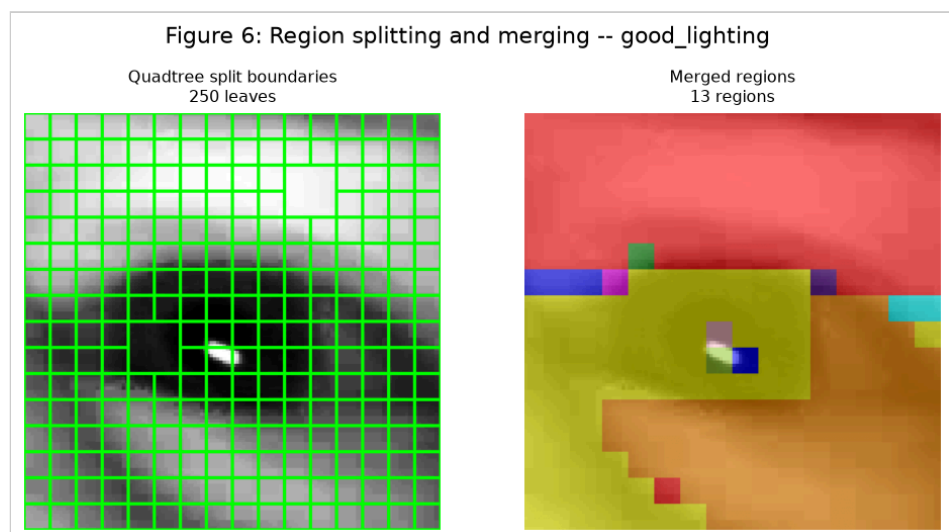


Figure 6: Quadtree split boundaries and final merged regions – good\_lighting.

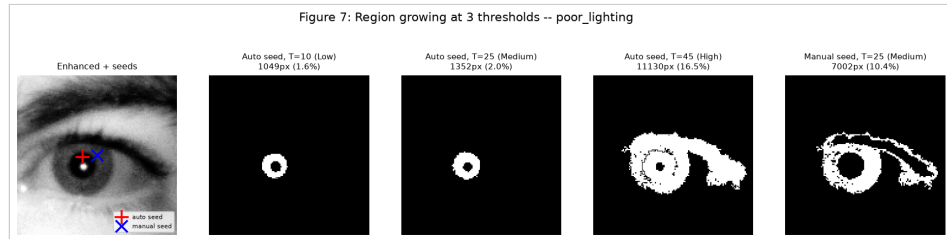


Figure 7: Region growing, auto seed at Low/Medium/High T, and manual seed at Medium T – poor\_lighting.

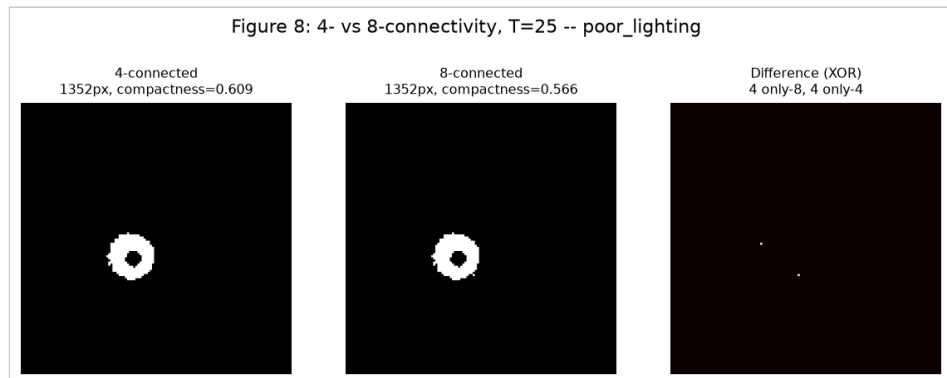


Figure 8: 4- vs 8-connectivity at T=25, with XOR difference map – poor\_lighting.

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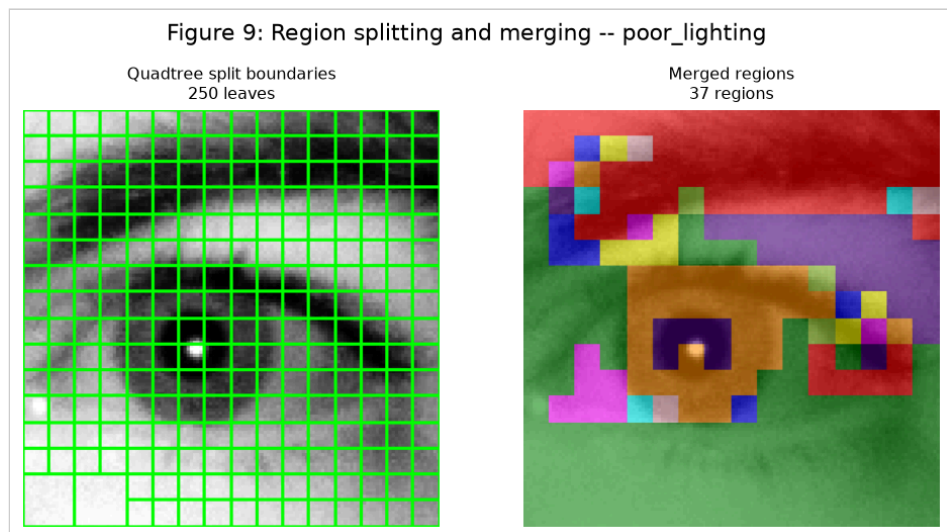


Figure 9: Quadtree split boundaries and final merged regions – poor\_lighting.

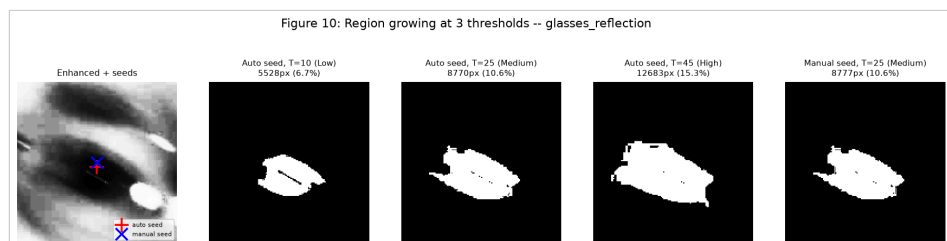


Figure 10: Region growing, auto seed at Low/Medium/High T, and manual seed at Medium T – glasses\_reflection.

Figure 11: 4- vs 8-connectivity, T=25 -- glasses\_reflection

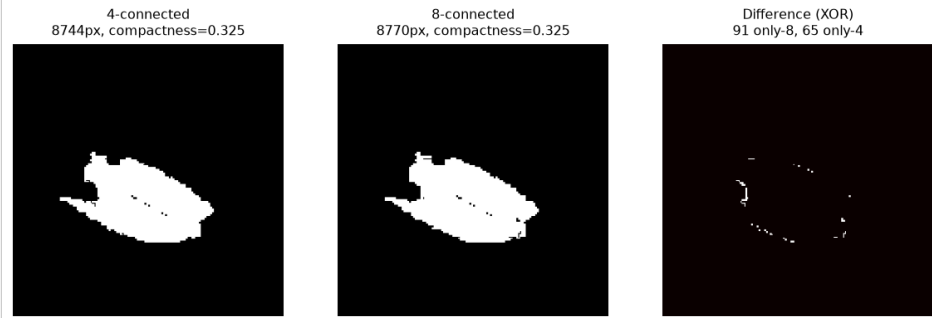


Figure 11: 4- vs 8-connectivity at T=25, with XOR difference map – glasses\_reflection.

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Figure 12: Region splitting and merging -- glasses\_reflection

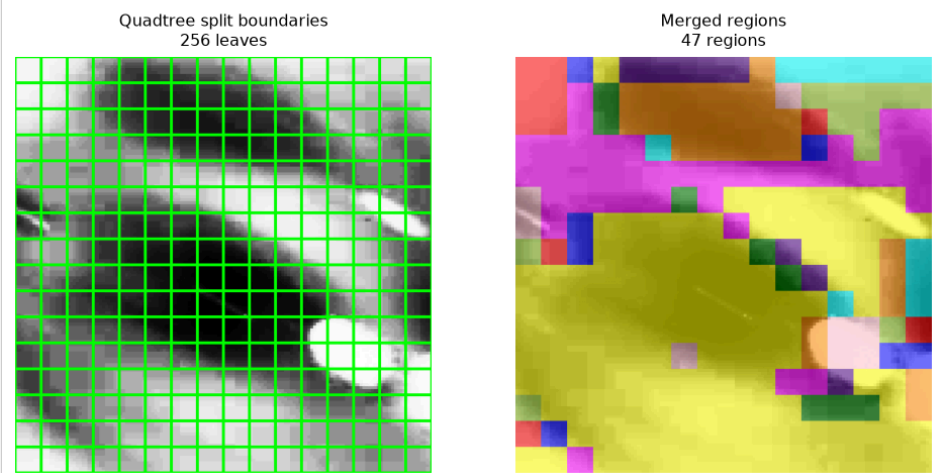


Figure 12: Quadtree split boundaries and final merged regions – glasses\_reflection.

Figure 13: Region-growing size vs. threshold, all 3 images

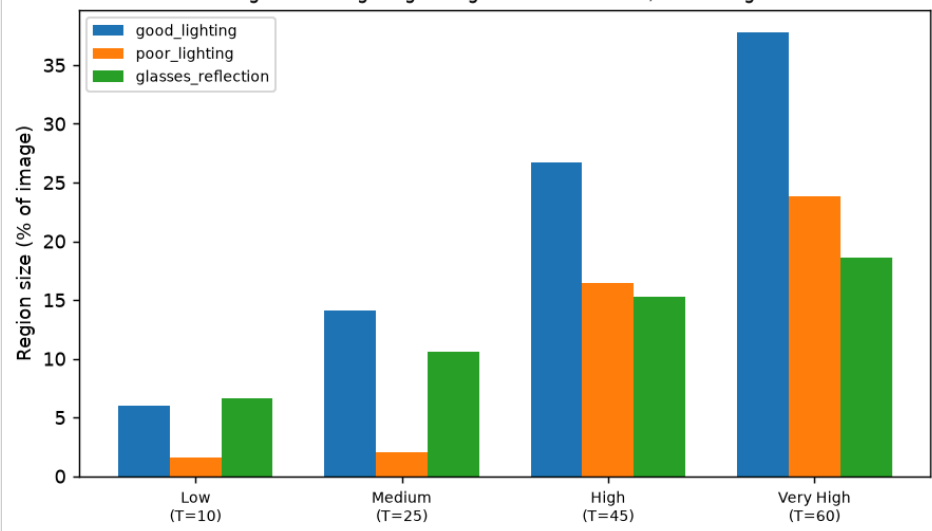


Figure 13: Region size (% of image) vs. threshold, all 3 images.

### Threshold effect (region size, missing pixels, leakage, segmentation accuracy)

"Missing pixels" / "leakage" / "segmentation accuracy" are measured against a reference pupil mask reused as-is from `src.segment.segment_eye` (Otsu value scaled by 0.7 to anchor on the pupil's darker core, then largest-connected-component isolation) – not a hand-labelled

ground truth, but the same reference Task 6 documents as this project's own pupil-isolation anchor. Missing = reference pixels the grown region has not yet reached; leakage = grown-region pixels outside that reference; accuracy = IoU between the two.

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| Image              | Thresh old       | T         | Region size (px, % image) | Reference pupil px | Missing px | Leakage px   | Accuracy (IoU) |
|--------------------|------------------|-----------|---------------------------|--------------------|------------|--------------|----------------|
| good_lighting      | Low              | 10        | 3,961 (6.0%)              | 22,316             | 18,355     | 0            | 0.178          |
|                    | Medium           | 25        | 9,246 (14.1%)             |                    | 13,070     | 0            | 0.414          |
|                    | High             | 45        | 17,523 (26.7%)            |                    | 4,793      | 0            | 0.785          |
|                    | <b>Very High</b> | <b>60</b> | <b>24,750 (37.8%)</b>     |                    | <b>429</b> | <b>2,863</b> | <b>0.869</b>   |
| poor_lighting      | Low              | 10        | 1,049 (1.6%)              | 11,880             | 10,831     | 0            | 0.088          |
|                    | Medium           | 25        | 1,352 (2.0%)              |                    | 10,528     | 0            | 0.114          |
|                    | <b>High</b>      | <b>45</b> | <b>11,130 (16.5%)</b>     |                    | <b>750</b> | <b>0</b>     | <b>0.937</b>   |
|                    | Very High        | 60        | 16,134 (23.9%)            |                    | 21         | <b>4,275</b> | 0.734          |
| glasses_reflection | Low              | 10        | 5,528 (6.7%)              | 15,776             | 10,248     | 0            | 0.350          |
|                    | Medium           | 25        | 8,770 (10.6%)             |                    | 7,006      | 0            | 0.556          |
|                    | High             | 45        | 12,683 (15.3%)            |                    | 3,093      | 0            | 0.804          |
|                    | <b>Very High</b> | <b>60</b> | <b>15,404 (18.6%)</b>     |                    | <b>372</b> | <b>0</b>     | <b>0.976</b>   |

#### 4- vs 8-connected comparison, T=25 (Medium)

| Connectivity       | Region size (px)             | Shape (compactness, mean)  | Connectivity of diagonal pixels            | Region leakage (extra px from diagonal steps) | Processing time (mean, ms/call) | Suitability for this application                                                                                                                                                           |
|--------------------|------------------------------|----------------------------|--------------------------------------------|-----------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4-connected        | 9,066 / 1,352 / 8,744        | 0.487 (rounder)            | Excluded – only orthogonal steps           | 0 (reference)                                 | 3.90                            | Cheaper, but misses diagonal-adjacent pupil-boundary pixels                                                                                                                                |
| <b>8-connected</b> | <b>9,246 / 1,352 / 8,770</b> | <b>0.435 (more jagged)</b> | <b>Included – orthogonal + 4 diagonals</b> | <b>+252 / +4 / +91 px (up to 2.8%)</b>        | <b>5.68 (1.45x slower)</b>      | <b>Recommended default – absorbs real diagonal-adjacent pupil pixels that 4-conn's orthogonal-only steps miss; ~46% slower per call is immaterial at &lt;10ms/call on these crop sizes</b> |

Region size triplets are (good\_lighting / poor\_lighting / glasses\_reflection). On poor\_lighting the two connectivities give an **identical** pixel count (1,352) – the small, tightly-bounded pupil ring there never needs a diagonal-only step to reach any of its neighbours. Neither connectivity is strictly a superset of the other: on good\_lighting, 252 pixels are reachable only via 8-connectivity's diagonal steps, but 72 pixels are absorbed by 4-connectivity *and not* 8-connectivity – because 8-connectivity's diagonal-first paths shift the

running region mean earlier in the BFS, so a handful of borderline pixels that 4-connectivity's slower orthogonal-only mean drift still admits fail the 8-connected run's already-shifted mean test. Full per-image numbers in `results/task7_connectivity_metrics.csv`.

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### Split-merge (quadtree split, min block 20px, homogeneity/merge tolerance 25)

| Image              | Leaves after split | Regions after merge | Reduction |
|--------------------|--------------------|---------------------|-----------|
| good_lighting      | 250                | 13                  | 19.2x     |
| poor_lighting      | 250                | 37                  | 6.8x      |
| glasses_reflection | 256                | 47                  | 5.4x      |

### Observations

**Region growing and thresholds.** Region size grows monotonically with  $T$  on all 3 images, but not linearly: on `poor_lighting` the region barely moves from Low to Medium (1,049  $\rightarrow$  1,352px, a 1.3x step for a 2.5x change in  $T$ ) and then jumps 8.2x from Medium to High (1,352  $\rightarrow$  11,130px) – the pupil ring stays isolated until  $T$  crosses the intensity gap to the surrounding dark eyelid crease, at which point growth escapes the ring almost immediately (visible in Figure 7). `good_lighting` and `glasses_reflection` grow more smoothly (largest single-step ratios 2.33x and 1.59x) because their pupil-to-surround intensity gap is shallower. **Leakage past the reference pupil mask is zero at Low/Medium/High on every image** and only appears at Very High ( $T=60$ ): 2,863px on `good_lighting`, 4,275px on `poor_lighting`, but still 0px on `glasses_reflection` – the IR glare hotspot's sharp bright/dark boundary in that image acts as a natural stopping edge that constrains growth even at the highest threshold tested, exactly the boundary-preserving behaviour Figure 10 shows (the mask hugs the eyelid contour without spilling past it at any of the 4 thresholds). Leakage does not automatically hurt accuracy: on `good_lighting` and `glasses_reflection`, IoU keeps rising through Very High (0.785 $\rightarrow$ 0.869 and 0.804 $\rightarrow$ 0.976) because missing pixels keep shrinking faster than leakage grows, but on `poor_lighting` IoU actually peaks at High (0.937) and falls back to 0.734 at Very High – there, leakage growth (0 $\rightarrow$ 4,275px) outpaces the remaining recovery of missing pixels (750 $\rightarrow$ 21px), so continuing to raise  $T$  past that point actively hurts accuracy rather than helping it.

**Manual vs. automatic seeding.** The automatic seed (darkest point after a light blur) lands on a true 0-intensity pixel inside the pupil on all 3 images. The manual seed (crop centre) is intensity 1 on `good_lighting` and intensity 3 on `glasses_reflection` – close enough that its Medium-threshold region size is within 1% of the auto-seed result on both (e.g. `glasses_reflection`: 8,777px manual vs. 8,770px auto). But on `poor_lighting` the manual seed sits on an intensity-54 pixel (the pupil is measurably off-centre in that crop) and its Medium- $T$  region (7,002px) is over 5x larger than the auto-seed result (1,352px), because it starts from a brighter running mean and immediately incorporates part of the iris/sclera boundary that the correctly-centred auto seed does not yet reach at the same  $T$  (Figure 7, rightmost panel). Automatic darkest-point seeding is measurably more reliable here than assuming the pupil is centred.

**4- vs 8-connectivity.** The size difference is small in absolute terms (0-252px, at most 2.8% of the 4-connected count) but not zero, and not one-directional: 8-connectivity picks up 252/4/91 diagonal-reachable pixels the 4-connected run misses on the 3 images respectively, while 4-connectivity separately keeps 72/4/65 pixels that the 8-connected run's earlier diagonal absorptions end up excluding once its running mean has shifted – visible as both colours of speckle in the Figure 5/8/11 XOR maps, not a one-sided fringe. Shape compactness is measurably lower for 8-connectivity on 2 of 3 images (0.526  $\rightarrow$  0.415 on `good_lighting`, 0.609  $\rightarrow$  0.566 on `poor_lighting`; unchanged at 0.325 on `glasses_reflection`), i.e. the extra diagonal reach makes the boundary slightly more jagged rather than rounder. Processing time is consistently higher for 8-connectivity (1.42x-1.47x, mean 1.45x, across 30 timed repeats per image) – expected, since every popped

pixel tests 8 candidate neighbours instead of 4 – but the absolute cost difference (0.4–2.6ms/call) is irrelevant at this image scale.

**Region splitting and merging.** Splitting alone produces 250–256 leaves on all 3 images (nearly all blocks split down to the 20px minimum rather than stopping early on homogeneity – histogram equalization raises local contrast/noise enough that few 20px+ blocks satisfy the max-min  $\leq 25$  test before hitting the size floor). Merging collapses this by 5.4x–19.2x: most aggressively on `good_lighting` (250  $\rightarrow$  13 regions), whose large, evenly-lit sclera/skin background merges into very few big regions, and least on `glasses_reflection` (256  $\rightarrow$  47 regions), whose glare hotspot, eyelid crease and reflection streak fragment the intensity landscape into many small mean-distinct patches that never satisfy the merge tolerance with their neighbours (visible in Figure 12's much more fragmented colour map compared to Figure 6's).

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**Honest limitation.** The reference pupil mask used for missing/leakage/accuracy (`src.segment.segment_eye`'s Otsu-anchored dark-core mask) is itself an approximation, not ground truth – Task 6 already documents that dataset-wide Otsu-derived masks on these crops can include eyebrow/eyelash strips alongside the pupil. This is visible directly in Figure 7 (`poor_lighting`): the region-growing mask at High threshold visually connects into the eyelid crease above the pupil, yet the leakage column still reads 0 at that threshold, because the reference mask's own dark-core envelope already extends into that same crease. Accuracy numbers here should be read as "recovery of the same dark-core envelope Task 6 anchors on", not as pupil-only ground-truth accuracy.

## Conclusion

Seeded region growing with a continuously-updated region mean was implemented in `src/region_growing.py` and evaluated on the same 3 dataset images used in Tasks 3 and 6, using an automatic darkest-point seed (verified more reliable than a naive centred manual seed, especially on the off-centre pupil in `poor_lighting`) and 4 threshold levels. Region size grows with threshold on every image but the growth is non-linear, with a measured leakage-onset point (past the reference Otsu-anchored pupil mask) between the High and Very High thresholds on `good_lighting/poor_lighting`, and no leakage at all within the tested range on `glasses_reflection`, where the IR glare boundary constrains growth. **8-connectivity** is the better default for this application: it recovers a small but real set of diagonal- adjacent pupil-boundary pixels (up to 2.8% more coverage) that 4-connectivity's orthogonal-only steps miss, at a processing-time cost (mean 1.45x slower, low single-digit milliseconds either way) that is immaterial at this image scale – though the comparison also showed neither connectivity is strictly a superset of the other, since the running-mean update makes the two BFS traversals diverge slightly rather than merely differ by a fixed diagonal fringe. **Region splitting and merging** (quadtree split + union-find merge) was implemented as a second region-based technique in the same module: splitting alone produced 250–256 leaves on all 3 images (most blocks hit the 20px size floor rather than stopping early on homogeneity), and merging reduced this by 5.4x–19.2x, least aggressively on `glasses_reflection` whose glare/reflection/eyelid-crease intensity landscape resists merging into large uniform regions. All quantitative claims in this report are grounded in the measured CSVs (`results/task7_threshold_metrics.csv`, `results/task7_connectivity_metrics.csv`, `results/task7_splitmerge_metrics.csv`), not visual impression alone.